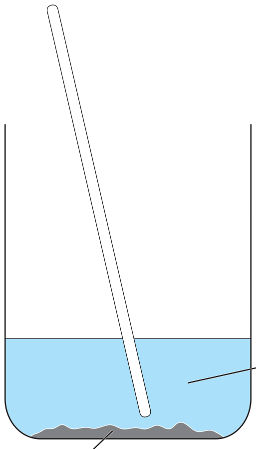
Examiner only

7. (a) A student investigated the mass of copper formed when magnesium powder was added to excess copper(II) sulfate solution.



The student added four different masses of magnesium powder to separate 50 cm^3 samples of copper(II) sulfate solution.

The table shows the results obtained.

Mass of magnesium added (g)	Mass of copper formed (g)
0.00	0.00
0.05	0.13
0.10	0.26
0.15	0.39
0.20	0.52

(i) Give **one** observation that would show that copper(II) sulfate is always in excess in this investigation. [1]

....

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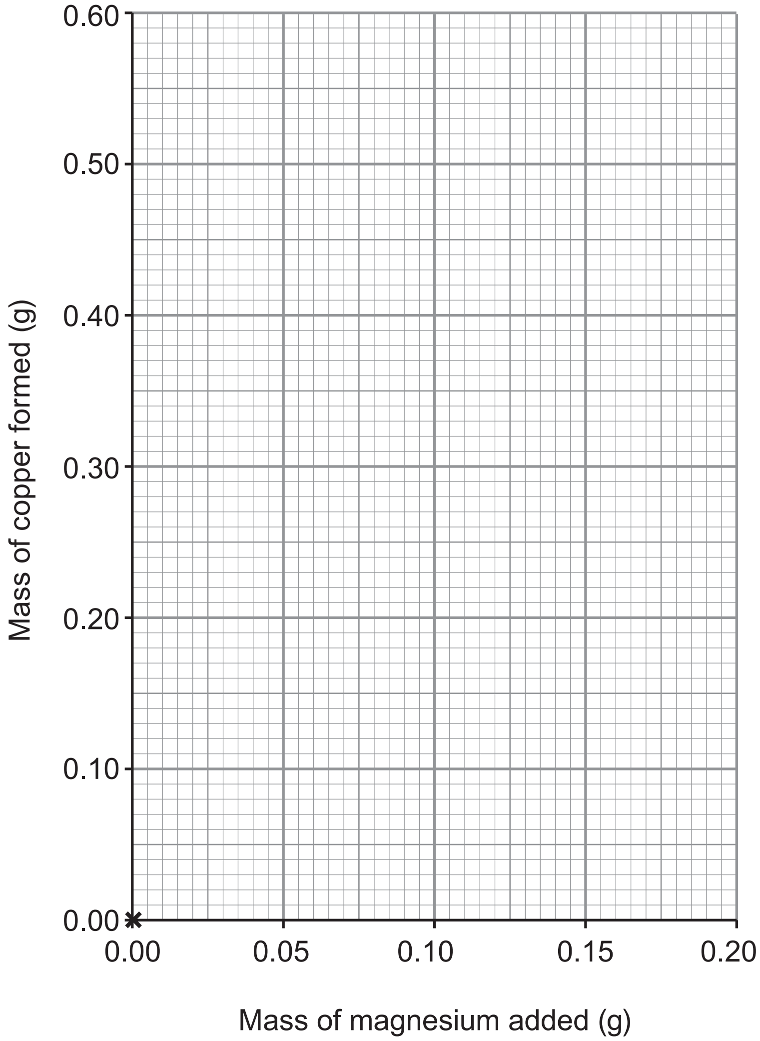
(ii) Plot the results from the table on the grid and draw a suitable line. [3]

The point (0,0) has been plotted for you.



(iii) I. Use the graph to find the mass of magnesium needed to form 0.18 g of copper. [1]

.... g

II. The student stated that 0.78 g of copper would be formed when 0.30 g of magnesium is used. Suggest how she arrived at this value. [1]

....

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(b) To ensure that the copper(II) sulfate solution was always in excess the technician dissolved 3.19 g of $CuSO_4$ in 500 cm^3 of water.

(i) Calculate the number of moles of copper(II) sulfate in 3.19 g. [2]

$A_r(O) = 16$ $A_r(S) = 32$ $A_r(Cu) = 63.5$

Number of moles = ... mol

(ii) Use the equation below to calculate the concentration of the copper(II) sulfate solution in mol/dm^3. [2]

$$\text{concentration (mol/dm}^3) = \frac{\text{number of moles}}{\text{volume (dm}^3)}$$

Concentration = ... mol/dm^3

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